

Problem List

Submit

0

Premium

Description

Editorial

Solutions

Submissions

977. Squares of a Sorted Array

EasyTopicsCompanies

Given an integer array `nums` sorted in **non-decreasing** order, return an array of **the squares of each number** sorted in non-decreasing order.

Example 1:

Input: `nums = [-4,-1,0,3,10]`
Output: `[0,1,9,16,100]`
Explanation: After squaring, the array becomes `[16,1,0,9,100]`. After sorting, it becomes `[0,1,9,16,100]`.

Example 2:

Input: `nums = [-7,-3,2,3,11]`
Output: `[4,9,9,49,121]`

Constraints:

- `1 <= nums.length <= 104`
- `-104 <= nums[i] <= 104`
- `nums` is sorted in **non-decreasing** order.

Follow up: Squaring each element and sorting the new array is very trivial, could you find an `O(n)` solution using a different approach?

Discover moreCoding Interview Prep

Seen this question in a real interview before? 1/6

Code

JavaAuto

```
1 class Solution {
2     public int[] sortedSquares(int[] nums) {
3         int n = nums.length;
4         int[] result = new int[n];
5
6         int left = 0;
7         int right = n - 1;
8
9         for (int i = n - 1; i >= 0; i--) {
10             int leftSquare = nums[left] * nums[left];
11             int rightSquare = nums[right] * nums[right];
12
13             if (leftSquare > rightSquare) {
14                 result[i] = leftSquare;
15                 left++;
16             } else {
17                 result[i] = rightSquare;
```

SavedLn 25, Col 1

Testcase

Test Result

AcceptedRuntime: 0 ms

Case 1Case 2

Input

nums =
[-4,-1,0,3,10]

Output

[0,1,9,16,100]

Expected

[0,1,9,16,100]